Ersatz Math Olymipad 2024 — P1/6

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There are 1001 stacks of coins $S_1, S_2, \ldots, S_{1001}$. Initially, stack S_k has k coins for each $k = 1, 2, \ldots, 1001$. In an operation, one selects an ordered pair (i, j) of indices i and j satisfying $1 \le i < j \le 1001$ subject to two conditions:

- The stacks S_i and S_j must each have at least one coin.
- The ordered pair (i, j) must not have been selected in any previous operation.

Then, if S_i and S_j have a coins and b coins, respectively, one removes $\gcd(a, b)$ coins from each stack.

What is the maximum number of times this operation could be performed?

Solution

The answer is 250500 times. Let $P(i, j)$ denote performing the given operation on the pair (i, j). Define the *n-shrink* process to be as follows. Let S_n be the stack with the most amount of coins. Perform the operations

$$P(n-1, n) \to P(n-2, n) \to P(n-3, n) \to \cdots$$

until we have performed $P(1, n)$ or the next operation would involve using a stack with zero coins.

Let $a < b$. A b-shrink and an a-shrink never repeat operations because the ordered pairs $(\_, b)$ and $(\_, a)$ are never the same since $a < b$. Suppose that the n-shrink process terminates after the operation $P(k, n)$, so the stack S_k has zero coins after the n-shrink. Then all of the stacks $S_{n-1}, S_{n-2}, \ldots, S_k$ have had one coin removed from them, and the stack S_n has had $n - k$ coins removed from it. Observe the following property of the n-shrink

$$(1, 2, 3, \ldots, n-2, n-1, n) \overset{n\text{-shrink}}{\longrightarrow} (0, 1, 2, \ldots, n-3, n-2, 1)$$

that is an n-shrink performed on $1, 2, \ldots, n$ always changes the last number to 1 and the first number to 0. Also we observe that if an n-shrink uses x operations, then the following $(n-1)$-shrink would use $x - 2$ operations.

Thus if we perform a 1001-shrink followed by a 1000-shrink followed and so on up to a 503-shrink the sequence (S_n) will look like so $0, 0, \ldots 0, 1, 2, 3, 1, 1, \ldots, 1$ where the zeros repeat 499 times and the ones repeat 499 times. This is because the after the 503-shrink we have performed $1001 - 503 + 1 = 499$ shrinks. Now after the 502-shrink the subsequence $1, 2, 3$ gets turned into $0, 1, 1$. And so now the overall sequence consists of 500 zeros, followed by 501 ones.

So in total we have performed 500 shrinks which means we have used the given operation $2 + 4 + \cdots + 1000 = 250500$ times. Now we just need to show that 250500 is in fact maximal.

If an operation $P(i, j)$ has $i \le 500$, then there can be at most $1 + 2 + \cdots + 500 = 500 \cdot 501/2$ of these operations since each operation will decrease the number of coins in stack S_i by at least 1. If an operation $P(i, j)$ has

$i > 500$ then there can be at most $\binom{501}{2} = 500 \cdot 501/2$ such operations since each pair (i, j) can be used once. Thus the upper bound for operations is $500 \cdot 501 = 250500$ as desired. ∎